Airborne SAR Suppression of Blanket Jamming Based on Second Order Blind Identification and Fractional Order Fourier Transform

Si Chen, *Senior Member, IEEE*, Yang Lin, Yue Yuan, *Member, IEEE*, Jianchao Li, Xun Wang, Lingzhi Zhu, *Member, IEEE*, and Shuning Zhang

***Abstract*— The presence of blanket jamming, a typical form of airborne synthetic aperture radar (SAR) jamming, causes incoherent signals with strong power to enter the airborne SAR receiver, which significantly reduces the signal-to-noise ratio (SNR) of airborne SAR images and dramatically influences the imaging effect of airborne SAR. In this manuscript, an airborne SAR anti-blanket interference algorithm based on fractional order Fourier transform (FRFT) and second-order blind identification (SOBI) is proposed. First, a geometrical model for airborne SAR imaging under a blanket interference condition is developed. Then, a blind signal separation (BSS) algorithm using SOBI is presented. Finally, to solve the problem of separation uncertainty of the existing BSS algorithms, the FRFT is used for signal identification. This method uses the SOBI algorithm to perform the BSS algorithm. Considering that FRFT can detect linear frequency modulation (LFM) signals emitted by airborne SAR, FRFT is used to solve the disadvantages of the current BSS methods. This algorithm separates signals with a high degree of similarity and performs well in terms of blanket jamming suppression. Simulation and measured experiments prove the validity of the algorithm.**

***Index Terms*— Blanket jamming suppression, blind signal separation (BSS), fractional order Fourier transform (FRFT), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave remote sensor that uses the principle of phase synthesis to obtain a 2-D distribution of electromagnetic scattering coefficients in the observation area [1], [2], [3], [4]. With its all-day, all-weather, and penetrating characteristics, SAR systems have unique advantages in military applications and are playing an increasingly prominent role in modern high-tech warfare [5], [6], [7]. Blanket jamming is an effective way to counter airborne SAR [8], [9], [10]. By transmitting the pulse or continuous wave that does not match the SAR signal, the blanket interference forms random or evenly distributed bright spots covering the entire SAR image after the airborne SAR processing, resulting in the unusable SAR image. Both noise amplitude modulation (NAM) jamming and noise frequency modulation (NFM) jamming are typical kinds of blanket interference. Both two types of interference covered the target signal by emitting a high-power jamming signal, causing the SAR image to become blurred and difficult to identify the ground target. Therefore, these two types of blanket jamming will be the focus of this manuscript.

Blind signal separation (BSS) is one signal processing technology in which the radiation source signal is recovered only according to the observed signal when the prior information of the target echo signal and transmission channel is unknown [11], [12], [13]. Its unique advantages in signal recovery have attracted extensive attention from scholars in the domains of speech processing, biomedical engineering, and digital communications. Zhou et al. [14] proposes an independent component analysis (ICA) method using joint approximate diagonalization eigenmatrices (JADE) and applied it to the SAR narrow-band jamming suppression problem. Tao et al. [15] use the fast ICA (FAST-ICA) technique for the narrow-band jamming suppression encountered by SAR.

A. Previous Work

As the electromagnetic environment of airborne SAR becomes more and more complex, the impact of blanket interference on airborne SAR becomes more and more significant. Research on blanket jamming suppression by airborne SAR is necessary. Current algorithms for the suppression of blanket interference in SAR fall into three main kinds: parametric algorithms, non-parametric algorithms, and semiparametric algorithms [8] as follows:

1) *Parametric Algorithms:* Parametric algorithms build the parametric model of the blanket interference and estimate the model parameters to estimate and extract the blanket jamming. Djukanovic et al. [16] and Djukanovic and Popovic [17] propose two types of higher-order fuzzy function estimation methods to estimate the phase parameters of disturbances using polynomial fitting. Yang et al. [18] use the iterative adaptive approach (IAA) and the short-time Fourier

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The authors are with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: chensi354@njust.edu.cn; linyang0316@163.com; yuanyue@njust.edu.cn; LiJianchao@njust.edu.cn; wangxun4697@njust.edu.cn; zhulingzhi@njust.edu.cn; zhangshuning@njust.edu.cn).

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transform (STFT) to estimate the isolated frequency of broadband jamming and then use orthogonal subspace projections to suppress the interference.

2) *Nonparametric Algorithms:* Nonparametric algorithms use the variance in intensity features between the blanket jamming and the actual target echo signal to construct adaptive filters or use signal subspace projections to filter out interference. Meyer et al. [19] designs adaptive detection to determine narrowband and wideband interference locations and trap filter widths to design a complete interference suppression process, which was greatly validated in measured fully polarized ALOS PALSAR data. Feng et al. [20] propose a subband spectral phase cancellation method that uses the variance between adjacent spectral subband signals to approach the role of a trap filter and applies it to real-world data.
3) *Semiparametric Algorithms:* The semiparametric algorithms use the optimized model to restrain the jamming while protecting the target echo signal in the optimized model. Huang et al. [21] take advantage of a non-convex model to increase the precision of the jamming constraint, and it is used as the basis of fast matrix decomposition technique [22] and tensor theory [23], which can precisely describe jamming problems and greatly reduce the complexity of calculation. Nguyen et al. [24] and Joy et al. [25] take advantage of robust principal component analysis (RPCA) to robustly combat narrowband radio frequency interference through low-rank recovery.

B. Contributions

In this manuscript, an airborne SAR anti-interference method using second-order blind identification (SOBI) and fractional order Fourier transform (FRFT) is developed for blanket jamming suppression. This algorithm proposes a BSS algorithm using SOBI. Considering the order uncertainty problem of BSS algorithms, a signal identification method using FRFT is proposed. For the case of airborne SAR emitting linear frequency modulation (LFM) pulse signals, FRFT is used to analyze the variance between the jamming and the true echo signal in the FRFT spectrum for signal identification, which makes up for the shortcomings of the proposed anti-jamming algorithm.

The main contributions of this manuscript can be concluded as follows:

1) This manuscript proposes a BSS method based on second-order blind discrimination for airborne SAR suppression of blanket jamming.
2) This manuscript proposes a signal identification method using FRFT. In response to the unclearness in the separation order of the BSS algorithm proposed above for suppression of blanket interference, this algorithm compensates for the above existing problems by using the difference in the FRFT domain between the jamming and the target source signal for signal identification.
3) For simulation experiment data and measured experiment SAR data (RADARSAT-1) acquired by Canadian remote sensing satellites, the method shows superiority over existing algorithms with better blanket interference suppression performance and better signal reproduction under various signal-to-interference ratio (SIR) conditions.

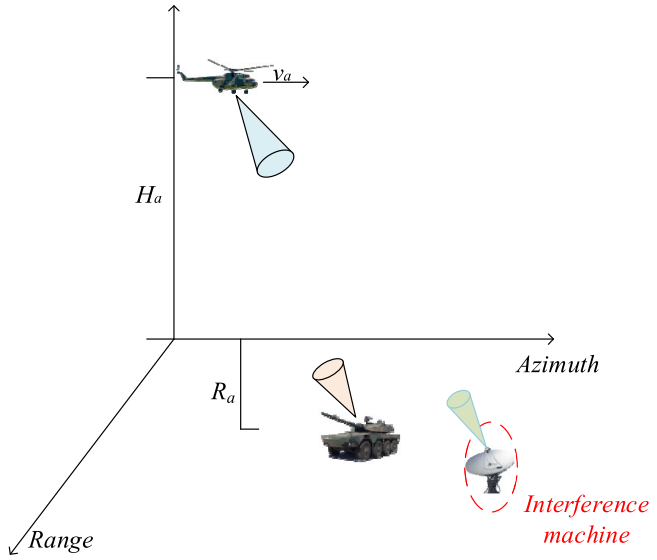


Fig. 1. SAR imaging theoretical model in a blanket jamming environment.

This manuscript is constructed as follows: In Section II, the airborne SAR imaging theoretical model and the hybrid model of the blanket jamming signal are developed. In Section III, the suppression of the blanket jamming algorithm of airborne SAR based on FRFT and SOBI is elaborated. In Section IV, the outcomes of the simulation and measured experiments are analyzed, and the validity of the presented algorithm is verified. In Section V, a brief conclusion of the whole manuscript is presented.

II. Related Work

A. Signal Model

Different imaging modes are used for different types of SAR. The common SAR imaging modes are strip mode, scan mode, and spotlight mode. This manuscript uses the classical range-Doppler (RD) imaging algorithm to perform airborne SAR imaging in strip SAR mode. SAR imaging theoretical model in a blanket interference condition is displayed in Fig. 1. The flight direction of the airborne SAR is the azimuth axis, and the flight speed of the airborne SAR is v_a. The altitude of the airborne SAR from the ground is H_a, and the projected distance of the ground target on the range axis is R_a. The shortest length between the ground target and the SAR platform occurs at the $t_a = t_c$ moment.

For airborne SAR systems, it is assumed that the airborne SAR launches an LFM pulse signal $s(t_r)$

$$s(t_r) = \text{rect}\left(\frac{t_r}{T_r}\right)\exp\left(j2\pi f_a t_r + j\pi k t_r^2\right) \tag{1}$$

where $\text{rect}(t_r/T_r) = \begin{cases} 1, & -T_r/2 \le t_r \le T_r/2 \\ 0, & \text{other} \end{cases}$, T_r is the transmitted signal pulsewidth, $k = B_r/T_r$ denotes the frequency

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modulation rate, B_r denotes the signal bandwidth, t_r denotes the pulse duration of the range axis, and f_a denotes the carrier frequency.

In each azimuth period t_m, the target echo signal received by the airborne SAR is $s(t_r, t_m)$

$$s(t_r, t_m) = \sigma \mathrm{rect}\left(\frac{t_r - \tau_r}{T}\right)\mathrm{rect}\left(\frac{t_m}{L_a/v}\right)\exp\left[j\pi k(t_r - \tau_r)^2\right] \times \exp(-j2\pi f_a \tau_r) \quad (2)$$

where σ is the scattering coefficient of the target, L_a denotes the synthetic aperture length, $\tau_r = 2R(t_m)/c$ is the delay time which is resulted from the distance from the ground target to airborne SAR, and c denotes the speed of light

$$R(t_a) = \sqrt{v_a^2(t_m - t_c)^2 + R_a^2 + H_a^2}. \quad (3)$$

B. Blanket Jamming Model

Blanket jamming can be divided into active and passive blanket interference. Active blanket interference uses a transmitter to emit a modulated jamming signal that saturates or overloads the receiving system of an enemy's electronic equipment to the point where no useful signal can be detected. Passive blanket jamming involves dropping foils that reflect electromagnetic waves over large areas of the air to create jamming corridors that mask your targets from enemy radar. The NAM jamming and the NFM jamming in active blanket interference are the focus of this manuscript.

The blanket jammer first generates a Gaussian white noise signal, then generates a noise amplitude-modulation signal, and finally generates a band-limited noise signal through a bandpass filter. The formula of NAM interference can be expressed as

$$J_1(t_r) = [U + U_1(t_r)]\cos\left(2\pi f_j t_r + \varphi_0\right) \quad (4)$$

where U denotes carrier amplitude, f_j is the center frequency of jamming, $U_1(t_r) = \exp(-t_r^2)/\sqrt{2\pi}$ is noise signal with zero mean, and the phase φ_0 is randomly distributed on the interval $[0, 2\pi]$.

The blanket jammer first generates a Gaussian white noise signal, then generates a noise frequency-modulation signal, and finally generates a band-limited noise signal through a bandpass filter. The formula of NFM jamming can be expressed as

$$J_2(t_r) = U\cos\left[\left(2\pi f_j t_r + 2\pi q\int_0^t U_2(t_r)dt_r\right) + \varphi_0\right] \quad (5)$$

where U denotes carrier amplitude, f_j denotes the center frequency of jamming, q denotes the frequency modulation rate, $U_2(t_r)$ denotes the noise signal with zero mean, and the phase φ_0 is of randomly distributed on the interval $[0, 2\pi]$.

C. Fractional Order Fourier Transform

Fourier analysis plays an extremely irreplaceable role in processing continuous-time and discrete-time signals. The traditional Fourier transform is an extremely effective tool in the field of analyzing and processing stationary signals but is weak when it comes to processing and analyzing non-stationary signals. To better analyze and process non-stationary signals, the FRFT is introduced.

The function $x(t_a)$ is defined in the time domain, whose p-order FRFT is a linear integration operation. It can be reflected as follows:

$$X_p(u) = \int_{-\infty}^{+\infty} K_p(u, t)x(t)dt \quad (6)$$

where $K_p(u, t)$ denotes the kernel function of FRFT, p is the FRFT order, $A_a = (1 - j\cot a)^{1/2}$, $a = p\pi/2$, $p = 2n$, and n is an integer number

$$K_p(u, t) = \begin{cases} A_a e^{j\pi(u^2\cot\alpha - 2ut\csc\alpha + t^2\cot a)}, & \alpha \neq n\pi \\ \delta(u - t), & \alpha = 2n\pi \\ \delta(u + t), & \alpha = (2n \pm 1)\pi. \end{cases} \quad (7)$$

With the variable substitution $u = u/\sqrt{2\pi}$ and $t = t/\sqrt{2\pi}$, (6) can be expressed as

$$\begin{aligned} X_p(u, t) &= \int_{-\infty}^{+\infty} K_p(u, t)x(t)dt, 0 < |p| < 2, 0 < |\alpha| < \pi \\ &= \begin{cases} \sqrt{\dfrac{1 - j\cot\alpha}{2\pi}} e^{j\frac{t^2+u^2}{2}\cot\alpha - \frac{jut}{\sin\alpha}} x(t)dt, & \alpha \neq n\pi \\ x(t), & \alpha = 2n\pi \\ x(-t), & \alpha = (2n \pm 1)\pi. \end{cases} \end{aligned} \quad (8)$$

III. Blanket Jamming Suppression Based on the BSS Algorithm Using FRFT and SOBI

As a research hotspot in signal processing in recent years, BSS is of great importance in many fields such as speech, communication, and biomedical engineering. In the chapter, a BSS algorithm using SOBI is presented. This method includes three processes: signal whitening, separation, and identification. In the signal whitening process, singular value decomposition (SVD) is used to derive the whitening matrix. In the signal separation process, the joint approximate diagonalization algorithm is proposed to derive the separation matrix to separate the signal which has been whitened. To solve the uncertainty disadvantages existing in the BSS method, we consider that airborne SAR transmits LFM pulse signals. Aiming at this feature, an LFM signal detection algorithm using FRFT is developed, which can identify the source signal from the separated jamming and the target echo signal. The separation order uncertainty disadvantage of the BSS method is solved well.

A. Linear Transient Mixing BSS Model

Signal mixing models for BSS can generally be classified into three types: linear transient mixing, linear convolutional mixing, and non-linear mixing. Linear transient mixing is the most mature model that has been studied and is also the simplest model. The process for a linear transient mixing

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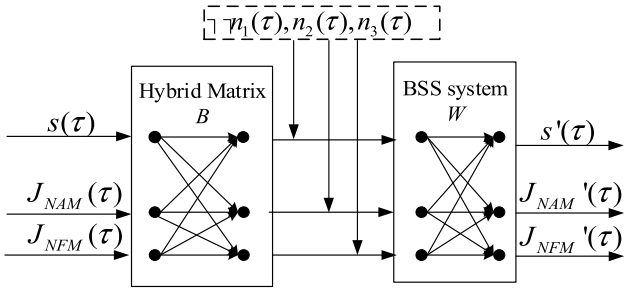


Fig. 2. Linear transient mixing BSS model.

model is displayed in Fig. 2. It is assumed that the target emits a signal of $s(\tau)$ and the jamming signal is $J(\tau)$

$$s(\tau) = [s_1(\tau), s_2(\tau), \ldots, s_n(\tau)] \tag{9}$$

$$J(\tau) = [J_1(\tau), J_2(\tau), \ldots, J_n(\tau)]. \tag{10}$$

The blanket interference signal and the true echo signal form the source signal $S(\tau)$

$$S(\tau) = [s(\tau), J(\tau)]^T \tag{11}$$

where $[\cdot]^T$ is the transpose operator.

The noise signal is superimposed on the actual echo signal in the process of transmission through the channel, and it is always there with or without a target echo. Following the linear transient mixing, the observed signal is $M(\tau)$:

$$M(\tau) = BS(\tau) + n(\tau) \tag{12}$$

where B is the hybrid matrix and $n(\tau)$ denotes the noise signal.

B. BSS Using SOBI

To ensure that the blind source separation algorithm proceeds smoothly, several conditions must be satisfied. First, the mixing matrix is a full column rank matrix, and the mixed signal can at most have one Gaussian signal [26]. Second, every source signal is a stochastic signal with a zero mean and is mutually freestanding. Last but not least, the observed signal consists of the actual echo signal, the jamming, and the noise, which are mutually freestanding.

The specific process of our proposed BSS algorithm consists of three processing: centering processing, whitening processing, and separating processing.

In order to satisfy the condition that every source signal is zero-mean, the observed signal $M(\tau)$ need to be centered. This processing method can be reflected as

$$M_p(\tau) = M(\tau) - \text{mean}[M(\tau)] \tag{13}$$

where $M_p(\tau)$ denotes the signal which has been centered and mean$[\cdot]$ represents the expectation arithmetic.

To eliminate variable coupling and satisfy the condition that each source signal is independent of the other, we need to perform the whitening processing on the centered observation signal $M_c(\tau)$. In this algorithm, the whitening processing is performed by means of SVD. The correlation matrix of the centered observation signal $M_c(\tau)$ is $R_x(s)$

$$R_x(s) = E\left[M_p(\tau)M_p^H(\tau)\right] = \frac{1}{N}\sum_{k=1}^{N} M(k)M^H(k-s) \tag{14}$$

where s represents the time delay. Next, we compute the zero-time delay correlation matrix $R_x(0)$ of $M_c(\tau)$ and perform SVD on it

$$R_x(0) = E\left[M_c(\tau)M_c^H(\tau)\right] = \frac{1}{N}\sum_{k=1}^{N} M(k)M^H(k) \tag{15}$$

$$R_x(0) = U\Lambda_s V^H \tag{16}$$

where Λ_s is the diagonal matrix. After SVD, the whitening matrix W can be represented as

$$W = \Lambda_s^{-\frac{1}{2}} U^H. \tag{17}$$

Then, joint diagonalization is used to derive the separation matrix Q. After whitening processing, the whitening signal $Y(\tau)$ can be obtained. The relation matrix of the whitening signal $Y(\tau)$ is $R_y(s)$

$$R_y(s) = E\left[Y(\tau)Y^H(\tau)\right] = \frac{1}{N}\sum_{k=1}^{N} Y(k)Y^H(k-s). \tag{18}$$

For the signal $Y(\tau)$, it is possible to use the joint approximate diagonalization algorithm to derive a unitary matrix Q so that the full array of delay covariance is simultaneously diagonalized [27]. Theorem 1 can be used to ensure the uniqueness of this unitary matrix Q.

Theorem 1 (Essential Uniqueness of Joint Diagonalization): Suppose the matrix $R_y(s) = [R_y(1), R_y(2), \ldots, R_y(N)]$, the matrix $R_y(s) = QD_sQ^T$, $1 \le s \le N$, Q is a unitary matrix, $D_s = [\lambda_1(s), \lambda_2(s), \ldots, \lambda_N(s)]$. The diagonalization operator Q of a matrix $R_y(s)$ is unique when and only when it satisfies: $d_i(s) \ne d_j(s), \forall 1 \le i \ne j \le N$.

The process of the joint approximate diagonalization can be expressed as

$$Q^T R_y(s) Q \approx D_s. \tag{19}$$

This process can be achieved using the minimization of non-negative functions $C(R_y(s), U)$

$$C(R_y(s), Q) = \sum_{k=1}^{N} \text{off}(Q^T R_y(s) Q) \tag{20}$$

where $\text{off}(Y) = \sum_{1 \le i \ne j \le n} |Y_{ij}|^2$. We use the Jacobi algorithm [45] to achieve joint diagonalization and use several Givens rotations to minimize $C(R_y(s), U)$. This process of getting the separation matrix Q can be expressed as

$$Q = \min_{Q} \sum_{i=1}^{N} \text{off}(Q^T R_y(s) Q). \tag{21}$$

And the separated signal is

$$X'(\tau) = QY(\tau) = QWM(\tau) = Q\Lambda_s^{-\frac{1}{2}} U^H M(\tau). \tag{22}$$

C. Signal Recognition Based on FRFT

There are many tools available for processing LFM pulse signals, such as STFT [28], [29], [30], Wigner vile distribution (WVD) [31], maximum likelihood estimation (MLE), etc., but they have disadvantages in the process of signal processing,

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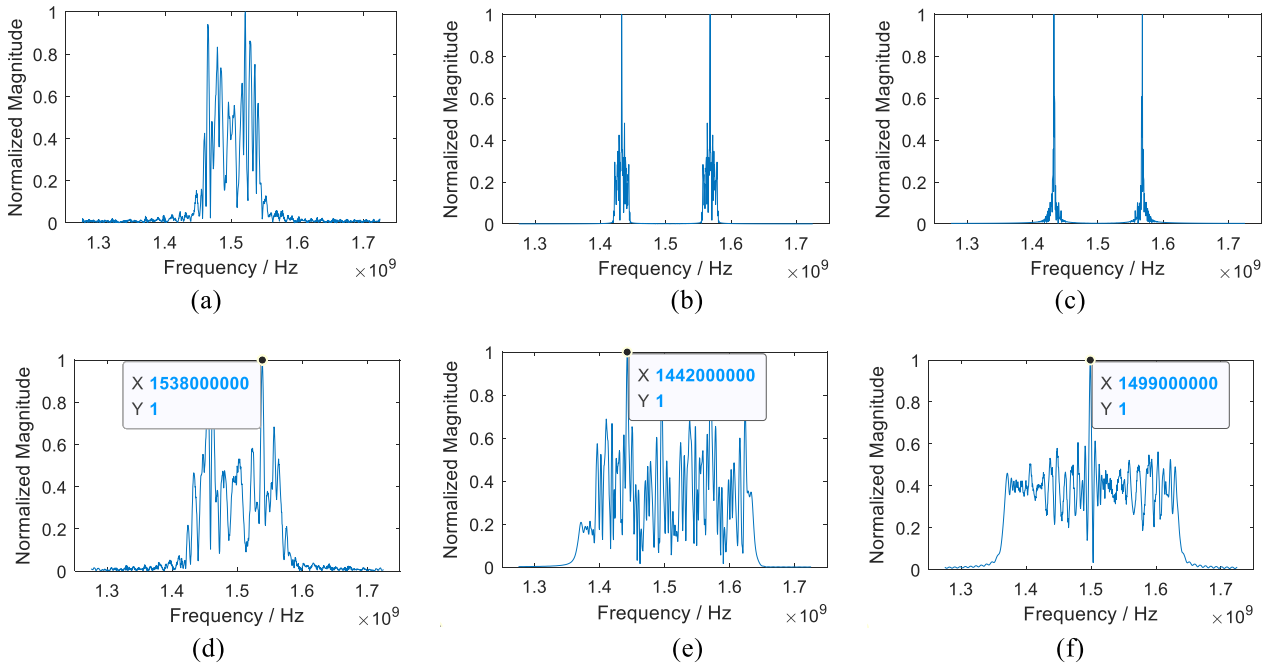


Fig. 3. Frequency and FRFT spectrums of the jamming signals and the single target echo signal. Frequency spectrum of the (a) single target signal, (b) NAM jamming signal, and (c) NFM jamming signal. FRFT spectrum of the (d) single target signal, (e) NAM jamming signal, and (f) NFM jamming signal.

such as the trade-off between time-frequency resolution and the vulnerability to cross-term jamming in the processing of multicomponent signals. FRFT can be understood as a representation of a signal in the time-frequency plane on a fractional domain consisting of an arbitrary angle of counterclockwise rotation around the origin, which creates a one-to-one correspondence between the time and frequency domains of the signal. Moreover, the LFM signal exhibits energy aggregation characteristics in a specific fractional domain [32], [33], [34].

Each azimuth signal of an airborne SAR receives an LFM pulse signal $s(t_r)$. The FRFT of the signal $s(t_r)$ can be reflected as

$$F_p(u) = A_p \exp^{j\pi u^2 \cot p} \times \int_{-T_r/2}^{T_r/2} \exp^{j\pi(\cot p + k)t_r^2} \exp^{j2\pi(f_c - u\csc p)t_r} dt_r \quad (23)$$

where A_p denotes the magnitude.

When $p = \operatorname{arc}\cot(-k)$, the FRFT amplitude of a finite length LFM pulse signal is a $\operatorname{sinc}(\cdot)$ function. It can be expressed as

$$F_p(u) = A_p T_r \frac{\sin[\pi(f_c - u\csc p)T_r]}{\pi(f_c - u\csc p)T_r}. \quad (24)$$

From (24), we can see that after the FRFT, the LFM pulse signal produces a maximum value at the moment $f_c - u\csc p = 0$. Where f_c is the starting frequency of the signal.

Tao et al. [15] identify the presence of narrowband interference in the echo by setting a spectral kurtosis threshold. Fig. 3 shows the frequency and FRFT spectrum for a single target echo signal and the interference signal, while Fig. 4 shows the frequency and FRFT spectrum for double target echo signal and the interference signal. Observing each of the images in Figs. 3 and 4, it is easy to find that the target identification using peak detection on the FRFT domain is feasible. Our proposed method has several advantages over the use of spectrum identification as follows:

1) Our proposed algorithm avoids the problem of misclassification caused by inappropriate threshold settings and is more reliable.
2) FRFT has better noise robustness, as the signal is affected by Gaussian white noise during channel transmission, the separated signal will also have some white noise, so FRFT can reduce the effect of white noise.
3) Recognition by extracting frequency spectrum features using classifiers is highly reliable but too computationally complex.

Therefore, the FRFT algorithm can be used to identify LFM pulse signals. Since in practice, the frequency modulation rate k of the target echo signal is known, we can take advantage of the equation $p = \operatorname{arc}\cot(-k)$ to obtain the best FRFT order that will detect the LFM pulse signal. Next, the equation $f_c - u\csc p = 0$ helps us to find the maximum value point of the LFM pulse signal on the FRFT spectrum.

D. Summary of Our Proposed Method

Based on the above discussion, the flowchart of the airborne SAR suppression of blanket jamming algorithm based on the BSS using FRFT and SOBI is shown in Fig. 5.

IV. Experimental Analysis

A BSS algorithm using SOBI and FRFT for blanket jamming suppression is developed in Section III. In this part, simulation and measured experiments are performed to verify the validity of our proposed method.

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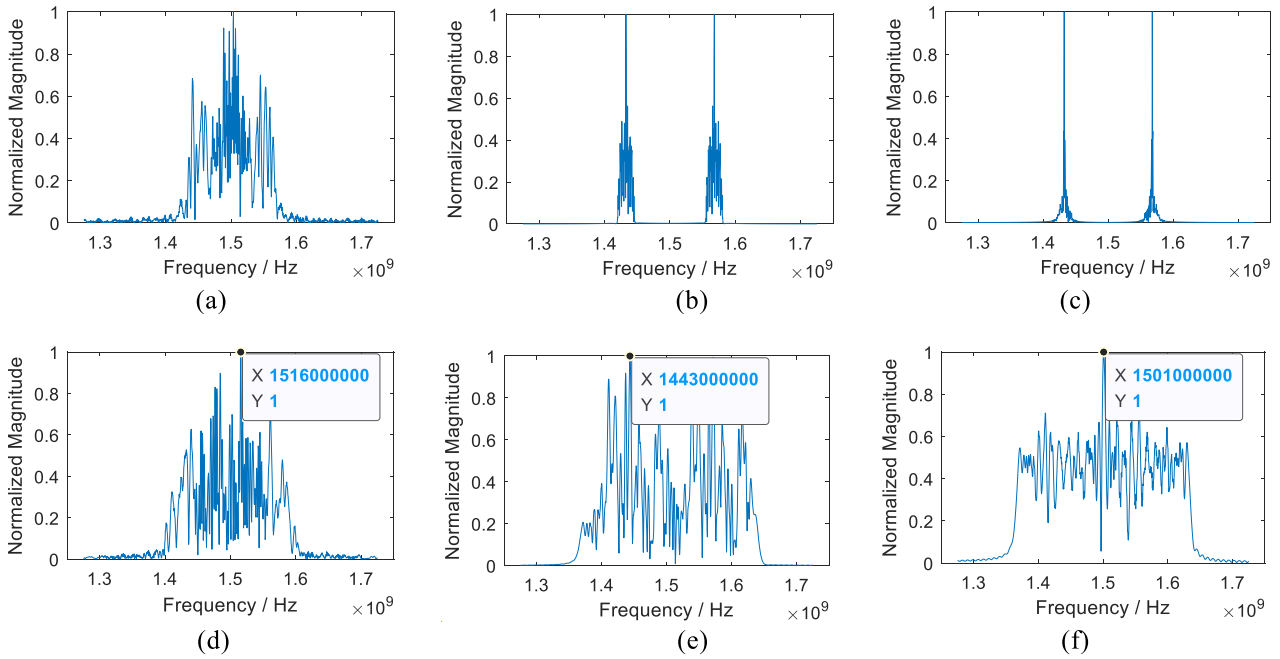


Fig. 4. Frequency and FRFT spectrums of the jamming signals and the double target echo signal. Frequency spectrum of the (a) double target signal, (b) NAM jamming signal, and (c) NFM jamming signal. FRFT spectrum of the (d) double target signal, (e) NAM jamming signal, and (f) NFM jamming signal.

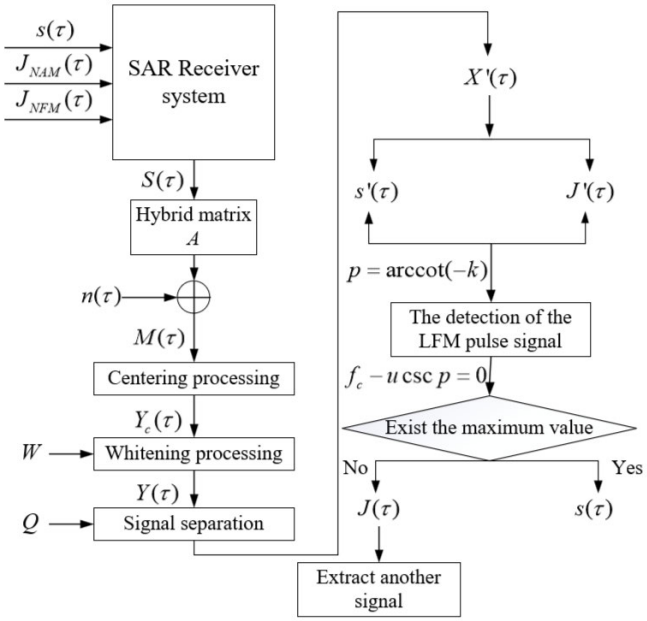


Fig. 5. Flowchart of the presented algorithm.

In this part, to demonstrate the interference suppression behavior of the presented algorithm, the adaptive Wiener filtering (AWF) algorithm proposed in [35] will be used as a comparison experiment.

A. Evaluation Indicators

To characterize the anti-jamming ability of the presented algorithm, two evaluation indicators are proposed to quantify the anti-jamming capability of this algorithm.

As a measure of signal similarity, the Pearson correlation coefficient (PCC) can be used to measure the similarity between the signal following blanket jamming suppression and the target echo signal. Assume that the signal after the suppression of blanket jamming is $s'(\tau)$. The PCC of the target source signal and the signal after the suppression of blanket jamming can be expressed as

$$\mathrm{PCC} = \frac{\sum_{n=1}^{Na*N_r} [s(\tau) - \overline{s(\tau)}][s'(\tau) - \overline{s'(\tau)}]}{\sqrt{\sum_{n=1}^{Na*N_r} [s(\tau) - \overline{s(\tau)}]^2} \sqrt{\sum_{n=1}^{Na*N_r} [s'(\tau) - \overline{s'(\tau)}]^2}} \tag{25}$$

where N_a denotes the azimuth samples, N_r denotes the range samples, $\overline{s(\tau)}$ is the average value of the theoretical original echo signal, and $\overline{s'(\tau)}$ denotes the average value of the signal after the suppression of blanket jamming. The closer the value of PCC is to 1, the higher the similarity between the signal following the suppression of blanket jamming and the true echo signal, and the better the anti-jamming behavior of this algorithm.

The image mean square error (IMSE) can be used for the measurement of picture quality to calculate the variance between images prior to and following blanket jamming suppression. The IMSE of the image before blanket jamming suppression and after blanket jamming suppression can be expressed as

$$\mathrm{IMSE} = 10 \lg \left\{ \frac{1}{xy} \sum_{m=0}^{x-1} \sum_{n=0}^{y-1} ||P(m,n) - Q(m,n)||^2 \right\} \tag{26}$$

where x is the row amount of the pictures, y is the column amount of the pictures, $P(m,n)$ is the grayscale of the picture prior to blanket jamming suppression, and $Q(m,n)$

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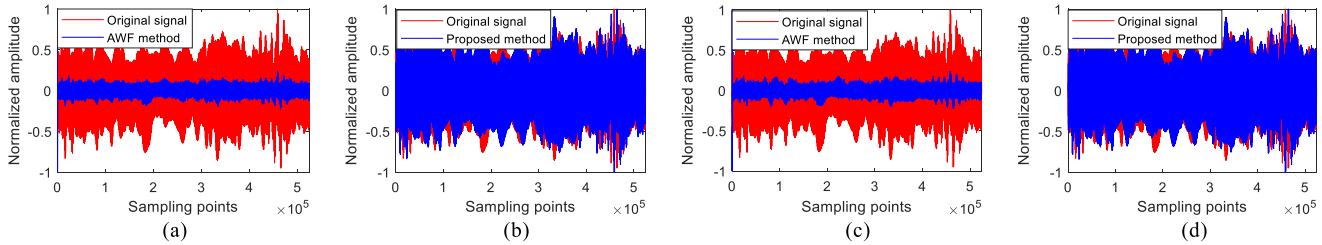


Fig. 6. Signal diagrams prior to and following NAM jamming suppression for single target simulation data. (a) AWF algorithm at the condition SIR $= -30$ dB. (b) Presented algorithm at the condition SIR $= -30$ dB. (c) AWF algorithm at the condition SIR $= -50$ dB. (d) Presented algorithm at the condition SIR $= -50$ dB.

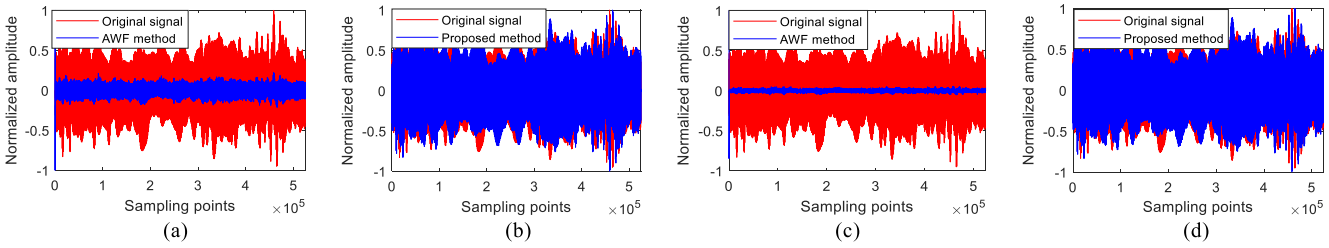


Fig. 7. Signal diagrams prior to and following NFM jamming suppression for single target simulation data. (a) AWF algorithm at the condition SIR $= -30$ dB. (b) Presented algorithm at the condition SIR $= -30$ dB. (c) AWF algorithm at the condition SIR $= -50$ dB. (d) Presented algorithm at the condition SIR $= -50$ dB.

TABLE I
SIMULATED SETTINGS

Settings	Size	Settings	Size
Central frequency	1.5GHz	Signal pulse width	1.5μs
Signal bandwidth	150MHz	Interference frequency	1.5GHz
Azimuth samples	1024	Interference bandwidth	450MHz
Range samples	512	The SNR of noise	10dB
SAR height	1000m	SAR speed	100m/s

is the grayscale of the picture following blanket jamming suppression. The smaller the value of IMSE, the greater the blanket interference suppression performance of the algorithm.

B. Single Target Simulation Experiments

The simulated parameters for the blanket jamming and airborne SAR targets are displayed in Table I. An aircraft is used as a simulated object. Then we add the NAM jamming signal and the NFM jamming signal with various SIR respectively. Finally, we describe the anti-jamming outcomes of the presented algorithm from both signal and image perspectives.

Fig. 6 reveals the signal diagrams for a single target prior to and following NAM jamming suppression. Viewing every picture in Fig. 6, it is easy to get the conclusion that our proposed algorithm has better robustness compared to the AWF algorithm at various SIR conditions. The signal after NAM interference suppression using our proposed algorithm is highly similar to the original signal with low signal loss. Different from our proposed algorithm, the signal processed by the AWF method has great signal loss compared with the actual echo signal and the signal distortion becomes more serious with the decrease of SIR. Fig. 7 reveals the signal diagrams for one single target prior to and following NFM jamming suppression. Viewing every picture in Fig. 7, we can get a similar conclusion.

Fig. 8 shows the single target simulation images prior to and following NAM jamming suppression. Through viewing every picture in Fig. 8, some findings can be obtained. At the condition SIR $= -30$ dB, we can find the outline of the object. At the condition SIR $= -50$ dB, the object is completely submerged in blanket jamming. Fig. 8(a) and (e) show the images after NAM jamming suppression using the AWF method. From the two figures, we can barely see the outline of the aircraft and there is still a clear area of interference in the airborne SAR image. Fig. 8(c) and (f) show the pictures after NAM jamming suppression using the presented algorithm. Compared with the image processed by the AWF algorithm, the image processed by the presented algorithm has better robustness and almost no jamming residue. Fig. 9 shows the single target simulation pictures prior to and following NFM jamming suppression. Viewing every picture in Fig. 9, similar conclusions can be obtained.

To quantify the jamming suppression ability of the presented algorithm, we use the evaluation indicators proposed above to perform a quantitative analysis. The PCC and the IMSE of the signals prior to and following blanket interference suppression are displayed in Table II. By observing the data in Table II, we further validate the conclusions reached earlier. Compared to the AWF method, our proposed algorithm has better robustness. Under strong interference conditions, the PCC of the signal before and after anti-interference can still be maintained at 0.99. And the IMSE of the presented algorithm is much smaller than the IMSE of the AWF algorithm.

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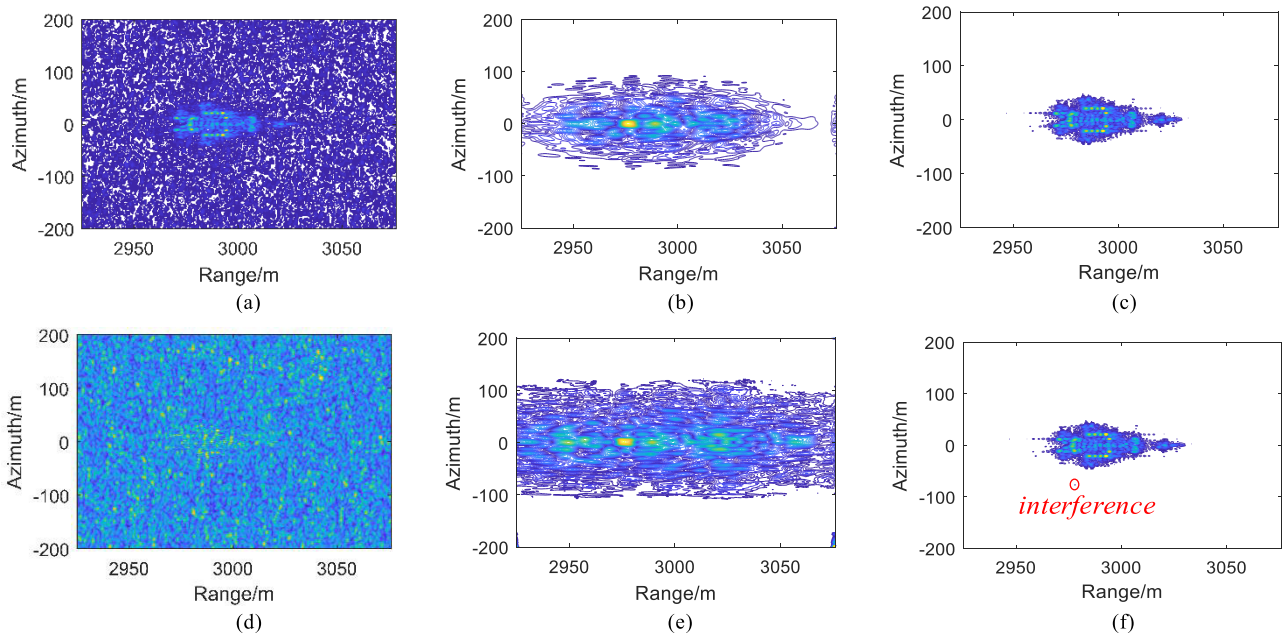


Fig. 8. Single target simulation images prior to and following NAM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR $= -30$ dB. (b) AWF algorithm jamming suppression picture at the condition SIR $= -30$ dB. (c) Presented algorithm jamming suppression picture at the condition SIR $= -30$ dB. (d) Picture prior to jamming suppression at the condition SIR $= -50$ dB. (e) AWF algorithm jamming suppression picture at the condition SIR $= -50$ dB. (f) Presented algorithm jamming suppression picture at the condition SIR $= -50$ dB.

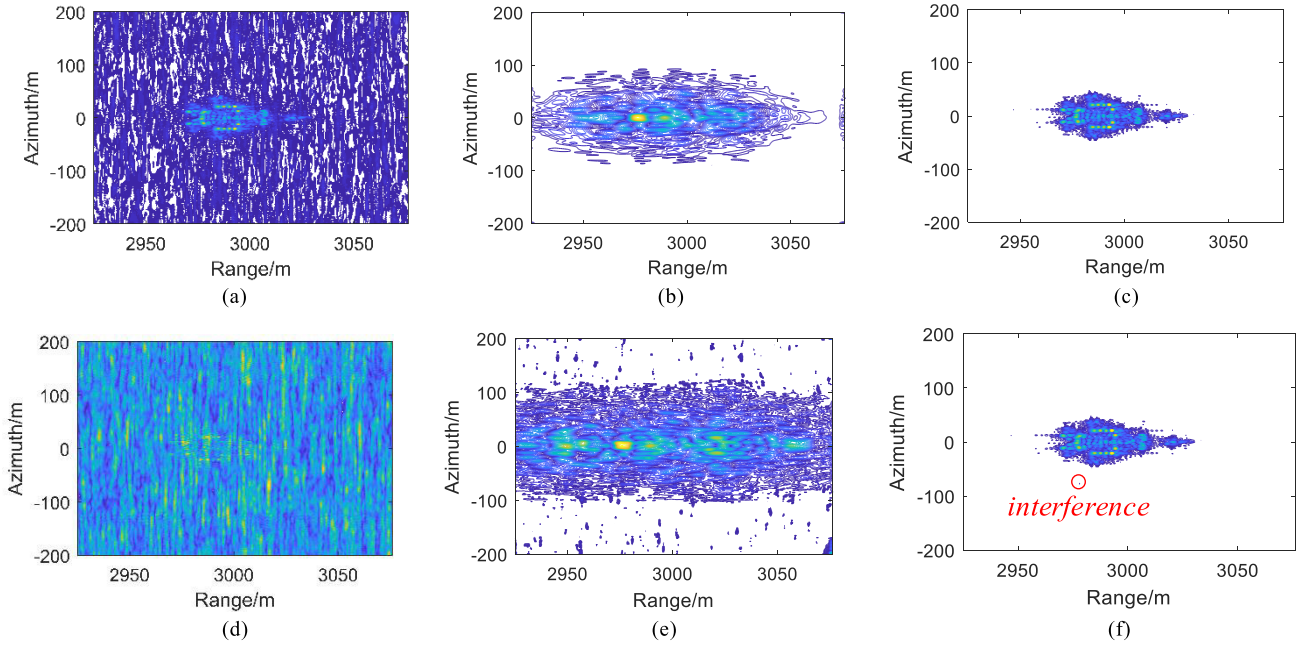


Fig. 9. Single target simulation images prior to and following NFM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR $= -30$ dB. (b) AWF algorithm jamming suppression picture at the condition SIR $= -30$ dB. (c) Presented algorithm jamming suppression picture at the condition SIR $= -30$ dB. (d) Picture prior to jamming suppression at the condition SIR $= -50$ dB. (e) AWF algorithm jamming suppression picture at the condition SIR $= -50$ dB. (f) Presented algorithm jamming suppression picture at the condition SIR $= -50$ dB.

C. Multiple Target Simulation Experiments

To make the simulation more complete, we supplemented the experiment with multiple targets as simulation objects.

Fig. 10 reveals the multiple target signal diagrams prior to and following NAM jamming suppression. Viewing every picture in Fig. 10, it is easy to get the conclusion that our proposed algorithm has better robustness compared to the AWF algorithm at various SIR conditions. The signal after NAM interference suppression using our proposed algorithm is highly similar to the original signal with low signal loss. Different from our proposed algorithm, the signal processed by the AWF method has great signal loss compared with the true echo signal and the signal distortion becomes more serious with the decrease of SIR. Fig. 11 reveals the signal diagrams prior to and following NFM jamming suppression.

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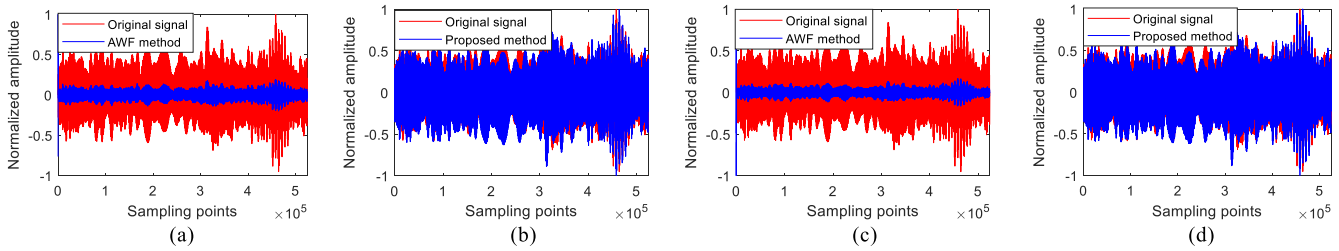


Fig. 10. Signal diagrams prior to and following NAM jamming suppression for multiple target simulation data. (a) AWF algorithm at the condition SIR $= -30$ dB. (b) Presented algorithm at the condition SIR $= -30$ dB. (c) AWF algorithm at the condition SIR $= -50$ dB. (d) Presented algorithm at the condition SIR $= -50$ dB.

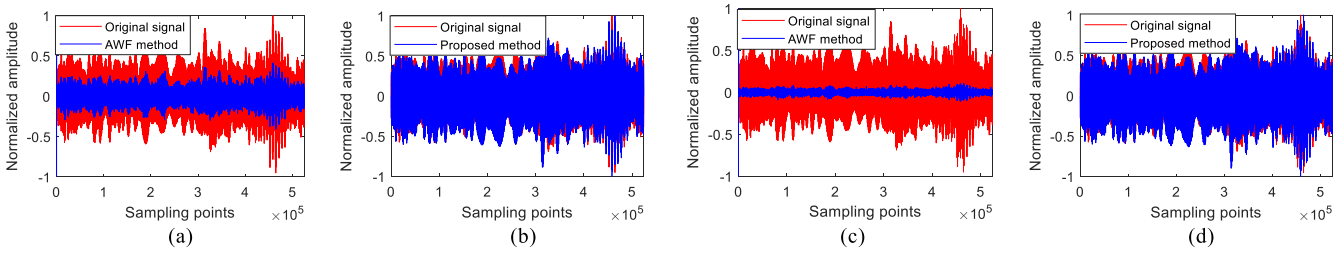


Fig. 11. Signal diagrams prior to and following NFM jamming suppression for multiple target simulation data. (a) AWF algorithm at the condition SIR $= -30$ dB. (b) Presented algorithm at the condition SIR $= -30$ dB. (c) AWF algorithm at the condition SIR $= -50$ dB. (d) Presented algorithm at the condition SIR $= -50$ dB.

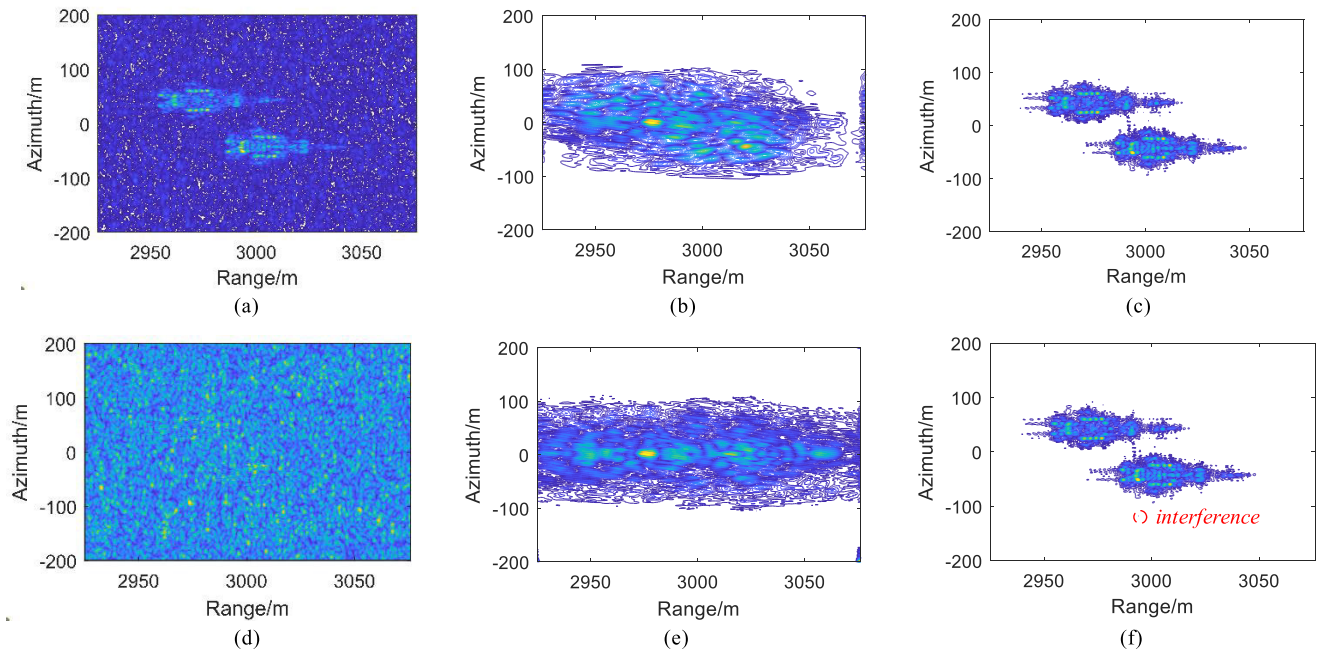


Fig. 12. Multiple target simulation images prior to and following NFM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR $= -30$ dB. (b) AWF algorithm jamming suppression picture at the condition SIR $= -30$ dB. (c) Presented algorithm jamming suppression picture at the condition SIR $= -30$ dB. (d) Picture prior to jamming suppression at the condition SIR $= -50$ dB. (e) AWF algorithm jamming suppression picture at the condition SIR $= -50$ dB. (f) Presented algorithm jamming suppression picture at the condition SIR $= -50$ dB.

Viewing every picture in Fig. 11, a similar conclusion can be achieved.

Fig. 12 shows the multiple target simulation images prior to and following NAM jamming suppression. Through viewing the pictures in Fig. 12, some findings can be obtained. Fig. 12(a) and (e) show the images after NAM jamming suppression using the AWF method. From the two figures, we can barely see the outline of the aircraft and there is still a clear area of interference on the airborne SAR image. When SJR $= -50$ dB, we cannot even see that there are two targets. Fig. 12(c) and (f) show the pictures after NAM jamming suppression using the presented algorithm. Compared with the image processed by the AWF algorithm, the image processed by the presented algorithm has better robustness and almost no jamming residue. Fig. 13 shows the simulation pictures prior to and following NFM jamming suppression.

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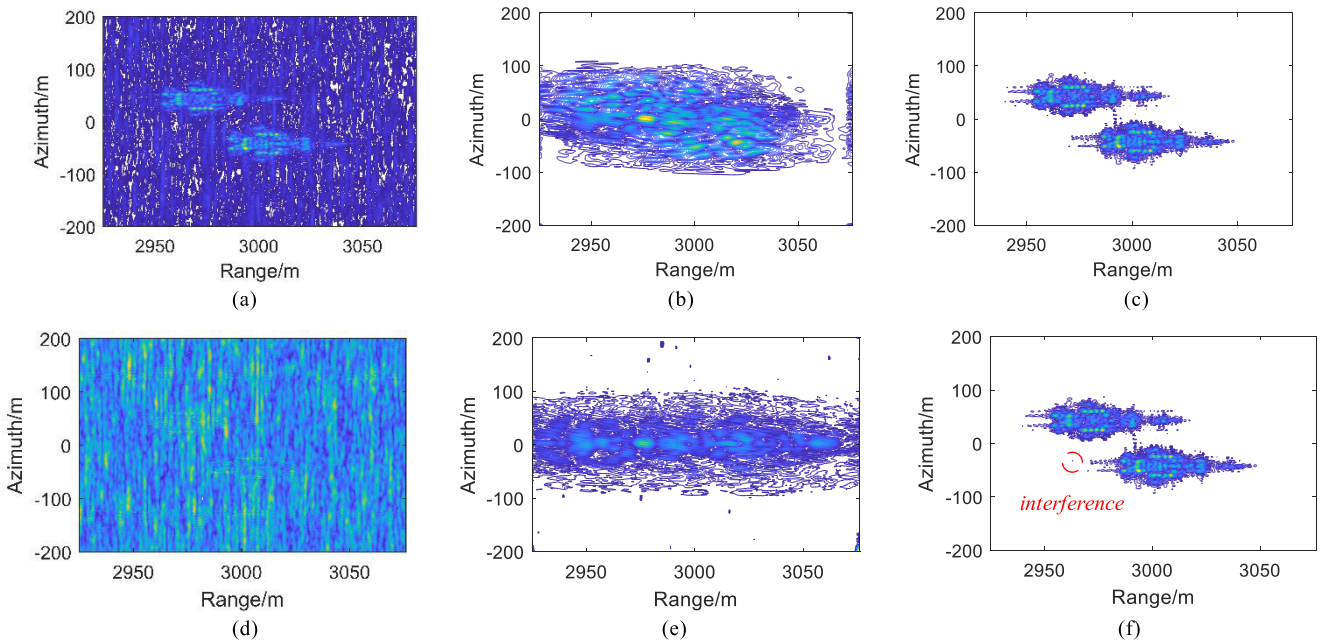


Fig. 13. Multiple target simulation images prior to and following NFM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR $= -30$ dB. (b) AWF algorithm jamming suppression picture at the condition SIR $= -30$ dB. (c) Presented algorithm jamming suppression picture at the condition SIR $= -30$ dB. (d) Picture prior to jamming suppression at the condition SIR $= -50$ dB. (e) AWF algorithm jamming suppression picture at the condition SIR $= -50$ dB. (f) Presented algorithm jamming suppression picture at the condition SIR $= -50$ dB.

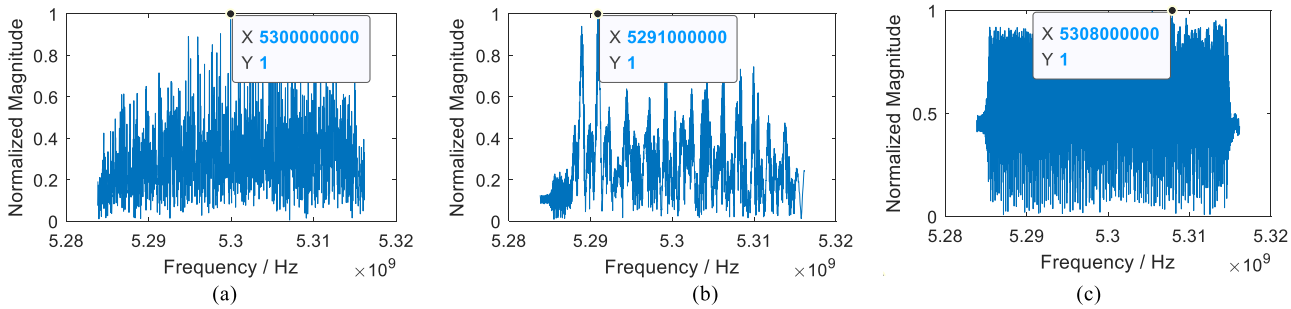


Fig. 14. FRFT spectrums of the jamming signals and Radarsat-1 signal. FRFT spectrum of the (a) Radarsat-1 signal t signal, (b) NAM jamming signal, and (c) NFM jamming signal.

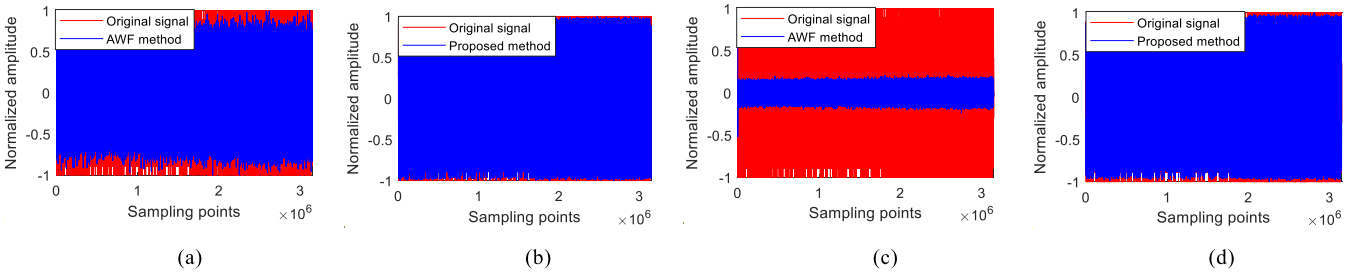


Fig. 15. Signal diagrams prior to and following NAM jamming suppression. (a) AWF algorithm at the condition SIR $= -15$ dB. (b) Presented algorithm at the condition SIR $= -15$ dB. (c) AWF algorithm at the condition SIR $= -35$ dB. (d) Presented algorithm at the condition SIR $= -35$ dB.

Viewing the pictures in Fig. 13, the same conclusions can be obtained.

To quantify the jamming suppression ability under multiple target conditions of the presented algorithm, we use the evaluation indicators proposed above to perform a quantitative analysis. The PCC and the IMSE of the signals prior to and following blanket interference suppression are displayed in Table III. After observing the data in Table III, we further validate the conclusions reached earlier. Compared to the AWF method, our proposed algorithm has better robustness. Under strong interference conditions, the PCC of the signal before and after anti-interference can still be maintained at 0.97. And the IMSE of the presented algorithm is much smaller than the IMSE of the AWF algorithm.

D. Measured Experiments

To further validate the validity of our presented algorithm, further experiments using the Radarsat-1 data launched by the Canadian Space Agency were carried out.

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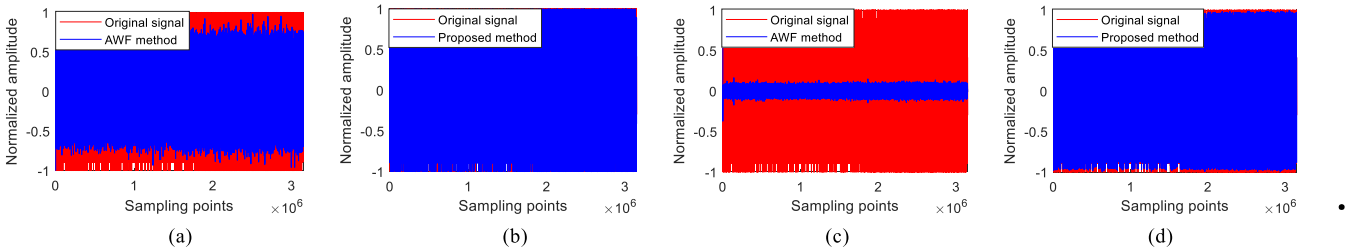


Fig. 16. Signal diagrams prior to and following NFM jamming suppression. (a) AWF algorithm at the condition SIR = −15 dB. (b) Presented algorithm at the condition SIR = −15 dB. (c) AWF algorithm at the condition SIR = −35 dB. (d) Presented algorithm at the condition SIR = −35 dB.

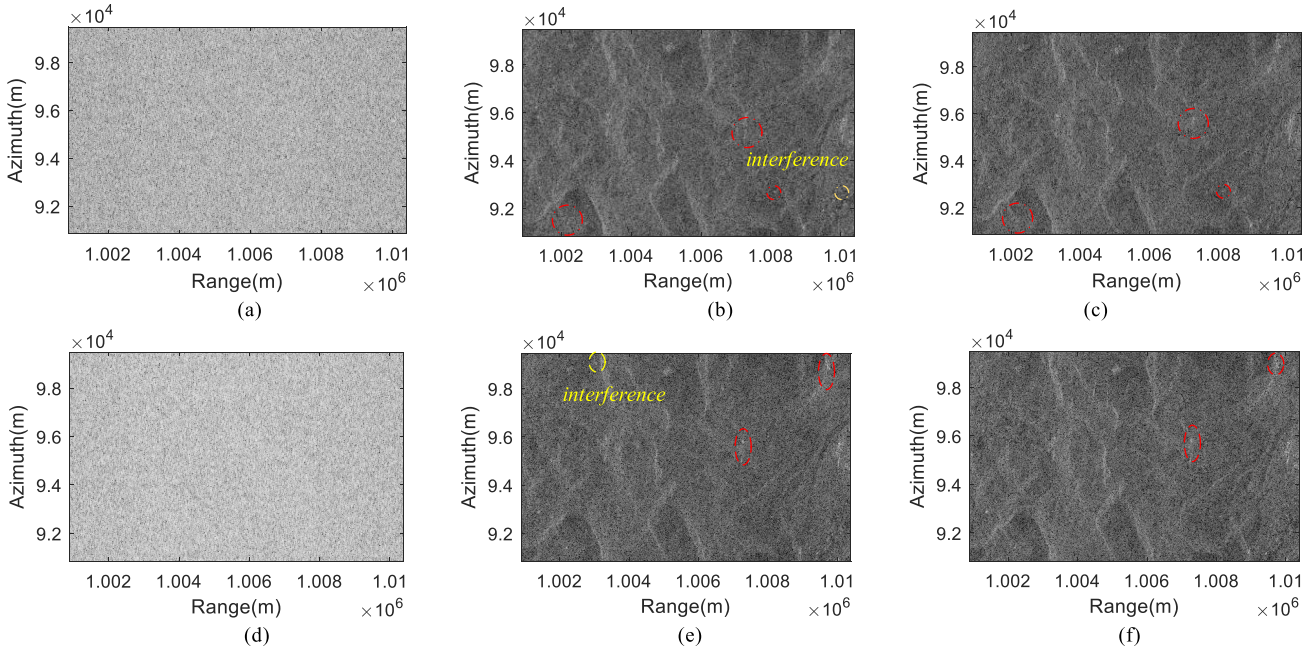


Fig. 17. Measured images prior to and following NAM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR = −15 dB. (b) AWF algorithm jamming suppression picture at the condition SIR = −15 dB. (c) Presented algorithm jamming suppression picture at the condition SIR = −15 dB. (d) Picture prior to jamming suppression at the condition SIR = −35 dB. (e) AWF algorithm jamming suppression picture at the condition SIR = −35 dB. (f) Presented algorithm jamming suppression picture at the condition SIR = −35 dB.

TABLE II
EVALUATION PARAMETERS FOR SINGLE TARGET SIMULATION DATA

NAM Jamming					
AWF Algorithm					
SIR (dB)	**-30**	**-35**	**-40**	**-45**	**-50**
PCC	0.7534	0.7408	0.7209	0.6901	0.6174
IMSE (dB)	-35.681	-36.013	-30.661	-24.83	-22.902
Our Presented Algorithm					
PCC	0.9999	0.9998	0.9993	0.9984	0.9978
IMSE (dB)	-50.974	-48.465	-44.581	-42.3	-41.774
NFM Jamming					
AWF Algorithm					
PCC	0.7854	0.7456	0.6679	0.5816	0.4785
IMSE (dB)	-35.461	-35.603	-31.708	-24.971	-28.054
Our Presented Algorithm					
PCC	0.9996	0.9995	0.9991	0.9986	0.9984
IMSE (dB)	-52.034	-47.163	-44.157	-41.855	-41.849

We put the NAM jamming and the NFM jamming from different SIRs into the Radarsat-1 data to investigate the anti-jamming performance of the presented algorithm. Fig. 14 shows FRFT spectrums of the jamming signals and Radarsat-1 signal. From Fig. 14, we can verify the conclusions presented above on the use of FRFT to detect LFM pulse signals. In the Radarsat-1 data, the FRFT algorithm mentioned above for the identification of the LFM pulse signal can still work well.

Fig. 15 shows the signal diagrams prior to and following NAM jamming. Observing each picture in Fig. 15, we can find that our proposed algorithm has better robustness compared to the AWF algorithm in various SIR situations. The signal after NAM interference suppression using our proposed algorithm is highly similar to the original signal with low signal loss. Different from our proposed algorithm, the signal processed by the AWF method has great signal loss compared with the actual echo signal, and signal distortion becomes more serious with the decrease of SIR. Fig. 16 reveals the signal waveforms prior to and following NFM jamming. By viewing the pictures in Fig. 16, the same conclusion can be obtained.

Fig. 17 shows the measured images prior to and following NAM jamming suppression. The imaging areas of Radarsat-1 data are Squamish town and Garibaldi Mountains. By viewing each picture in Fig. 17, some conclusions can be achieved. Fig. 17(a) and (e) show the images after NAM interference suppression using the AWF algorithm. Fig. 17(c) and (f) show

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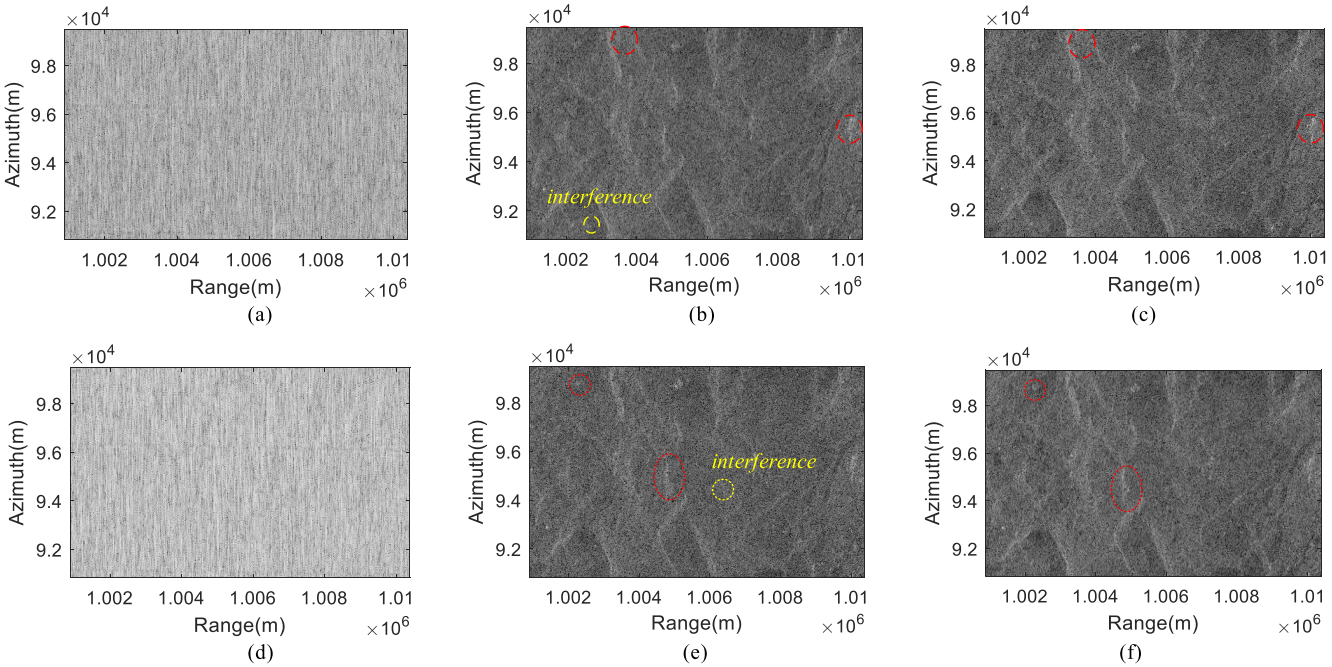


Fig. 18. Measured images prior to and following NFM jamming suppression. (a) Picture prior to jamming suppression at the condition SIR $= -15$ dB. (b) AWF algorithm jamming suppression picture at the condition SIR $= -15$ dB. (c) Presented algorithm jamming suppression picture at the condition SIR $= -15$ dB. (d) Picture prior to jamming suppression at the condition SIR $= -35$ dB. (e) AWF algorithm jamming suppression picture at the condition SIR $= -35$ dB. (f) Presented algorithm jamming suppression picture at the condition SIR $= -35$ dB.

TABLE III
EVALUATION PARAMETERS FOR MULTIPLE TARGET SIMULATION DATA

NAM Jamming					
AWF Algorithm					
SIR (dB)	**-30**	**-35**	**-40**	**-45**	**-50**
PCC	0.6531	0.6139	0.5623	0.4968	0.4722
IMSE (dB)	-33.132	-31.007	-25.345	-23.637	-22.624
Our Presented Algorithm					
PCC	0.9998	0.9985	0.9983	0.986	0.9788
IMSE (dB)	-41.507	-39.943	-36.025	-36.023	-35.981
NFM Jamming					
AWF Algorithm					
PCC	0.6832	0.6036	0.5193	0.4462	0.3622
IMSE (dB)	-32.658	-30.541	-25.933	-23.166	-24.737
Our Presented Algorithm					
PCC	0.9999	0.9997	0.9994	0.9991	0.9922
IMSE (dB)	-43.995	-39.336	-37.031	-36.191	-36.148

TABLE IV
EVALUATION PARAMETERS OF MEASURED TESTING

NAM Jamming					
AWF Algorithm					
SIR (dB)	**-15**	**-20**	**-25**	**-30**	**-35**
PCC	0.442	0.415	0.392	0.379	0.346
IMSE (dB)	-4.952	-4.820	-4.796	-4.795	-4.815
Our presented Algorithm					
PCC	0.998	0.998	0.997	0.996	0.989
IMSE (dB)	-4.977	-4.836	-4.819	-4.818	-4.836
NFM Jamming					
AWF Algorithm					
PCC	0.452	0.401	0.357	0.313	0.278
IMSE (dB)	-5.233	-4.874	-4.815	-4.808	-4.828
Our presented Algorithm					
PCC	0.998	0.998	0.997	0.983	0.975
IMSE (dB)	-5.234	-4.889	-4.84	-4.83	-4.846

the images after NAM jamming suppression using our presented algorithm. Due to the appearance of coherent patches, it is difficult to see in Fig. 17 that there exists a huge difference between our proposed method and the AWF method. However, we still find that our proposed algorithm reproduces the details of the mountain and the town better.

The mountains in the SAR image after NAM interference suppression by the AWF method still have some jamming spots, while the SAR image after NAM jamming suppression by our proposed algorithm has a much better reproduction of the mountains and the town than the AWF algorithm. By observing the area marked in red in Fig. 17, we can clearly see that our proposed algorithm is better at reproducing details. In the yellow marked area in Fig. 17, we can still find that the image still has interference points after the interference suppression of the AWF method. Fig. 18 shows the measured images prior to and following NFM jamming suppression. Viewing every picture in Fig. 18, similar conclusions can be obtained.

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To quantify the jamming suppression ability of the presented algorithm, the evaluation indicators proposed above are used to perform a quantitative analysis. The PCC and the IMSE of the signals prior to and following blanket interference suppression are displayed in Table IV. By observing the outcomes in Table IV, we further validate the conclusions reached earlier. Compared to the AWF algorithm, the presented algorithm has better robustness. Under strong jamming conditions, the PCC of the signal before and after anti-interference can still be maintained at 0.97. And the IMSE of the presented algorithm is smaller than the IMSE of the AWF algorithm.

V. Conclusion

In the manuscript, a BSS algorithm using SOBI and FRFT is presented to suppress blanket jamming for airborne SAR. First, the mixed signal model at the blanket jamming condition is proposed. Next, a BSS method using SOBI is presented, and the FRFT is performed to address the separation uncertainty disadvantage of the BSS method. The results of the single target simulated, multiple target simulated, and measured experiments demonstrate that our presented algorithm has a better performance on both PCC and IMSE with better jamming suppression performance compared to the AWF algorithm. Good robustness can also be proved in the presented algorithm.

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Si Chen (Senior Member, IEEE) was born in Xuancheng, Anhui, China, in February 1988. He received the B.S. and Ph.D. degrees in detection guidance technology, communication information engineering from the Nanjing University of Science and Technology, Nanjing, China, in 2010 and 2015, respectively.

He is currently an Associate Professor with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology. His research interests include missile-borne synthetic aperture radar (SAR) imaging, electronic countermeasures, and radar signal processing.



Yang Lin was born in Nantong, Jiangsu, China, in 2000. He received the B.S. degree in communication engineering from the Jinling Institute of Technology, Nanjing, China, in 2022. He is currently pursuing the M.S. degree in electronic and optical engineering with the Nanjing University of Science and Technology, Nanjing.

His research interests include synthetic aperture radar, signal processing, and target identification and recognition.



Yue Yuan (Member, IEEE) was born in Nanjing, Jiangsu, China, in May 1991. She received the B.S. and Ph.D. degrees in communication and information engineering from the Nanjing University of Science and Technology, Nanjing, China, in 2013 and 2020, respectively.

She is currently a Lecturer with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology. Her research interests include linear-array 3-D synthetic aperture radar (SAR) imaging and digital signal processing.



Jianchao Li was born in Anhui, China, in 1997. He received the B.S. degree in detection, guidance and control technology from the Nanjing University of Science and Technology, Nanjing, China, in 2020, where he is currently pursuing the Ph.D. degree in electronic and optical engineering.

His research interests include target identification and recognition.



Xun Wang received the B.S. degree in electronic information science and technology from the North China University of Water Resources and Electric Power, Zhengzhou, China, in 2019. He is currently pursuing the Ph.D. degree with the Nanjing University of Science and Technology, Nanjing, China.

His research interests include radar signal processing and anti-jamming.



Lingzhi Zhu (Member, IEEE) was born in Rugao, Jiangsu, China, in 1994. He received the B.S. degree in electronic information engineering and the Ph.D. degree in information and communication engineering from the Nanjing University of Science and Technology, Nanjing, China, in 2016 and 2021, respectively.

He holds a post-doctoral position in electronic science and technology with the Nanjing University of Science and Technology. His research interests include radar micro-Doppler, signal jamming and anti-jamming, target detection, and digital signal processing.



Shuning Zhang received the Ph.D. degree from the Nanjing University of Science and Technology, Nanjing, China, in 2006.

She is currently a Professor at the Laboratory of Radar Jamming and Anti-Jamming, Nanjing University of Science and Technology. She has authored or coauthored two books and more than 30 manuscripts. Her research interests include radar jamming and anti-jamming, signal identification, and target detection.

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